

Question 1 (15 Marks)	Marks
(a) Solve for x : $5^x \cdot 25^{2x+1} = 125^x$.	2
(b) Solve for m : $ m + 2 > 1 - m $	2
(c) Evaluate $\sqrt{2} \tan 60^\circ - \frac{\cos 30^\circ}{\sin 45^\circ}$	2
(d) If $x^2 + 3xy - 4y^2 = 0$, then the ratio $\frac{x}{y}$ has one value of unity. Find the other value.	1
(e) If $\tan A = -\frac{12}{5}$ and $\sin A > 0$, evaluate $\sec A$.	1
(f) Differentiate	
(i) $3\sqrt{\pi} - 2x^5$	1
(ii) $\sin x \cos x$	2
(iii) $\sqrt{(\ln 2x)^7}$	2
(iv) $\frac{2x}{4x+1}$	2

Question 2 (15 Marks) [Start a New Page]

(a) If $\log_{10} y = \frac{2}{3} \log_{10} x - 2$, express y in terms of x with no logarithmic expression.	3
(b) Find the gradient of the tangent to the curve $y = e^{x^2}$ when $x = 1$.	2
(c) $ABCDE$ is a regular pentagon. Diagonals BD and CE intersect at P .	
(i) With the aid of a diagram, show that $\triangle CDE \equiv \triangle BCD$.	3
(ii) Show that $PD = PC$.	2
(iii) Hence, or otherwise, show that $ABPE$ is a kite.	2
(d) Differentiate, by first principles the function, $f(x) = \sqrt{x+2}$.	3

Question 3 (15 Marks) [Start a New Page] **Marks**

- (a) Write the domain and range of $y = \ln(1 - x^2)$. **2**
- (b) Find the values of x for which the curve $y = x^3 - 12x$, is monotonically increasing. **2**
- (c) Find $\lim_{x \rightarrow 0} \frac{2 \sin x}{\tan\left(\frac{x}{2}\right)}$, giving reasons. **2**
- (d) Show that $y = \frac{x}{x^3 - x}$ an even function & state the restriction on x . **2**
- (e) If $\cos x = \frac{9 - a^2}{9 + a^2}$, find $\sin x$ in terms of a . **2**
- (f) Draw a neat sketch of $y = \frac{(x+1)^2}{(x-1)(x+2)}$, clearly showing all important features (calculus is not required). **3**
- (g) Find the point P that divides the interval AB internally in the ratio 5:3, if $A = (3, -7)$ and $B = (8, 2)$. **2**

Question 4 (15 Marks) [Start a New Page] **Marks**

- (a) Find the first 2 positive values of x such that $\sin 2x = 0.1$. . **2**
(Answer to 1 decimal place).
- (b) (i) Show that the locus of a point P such that $AP \perp BP$, where A and B are **2**
 $(1, 2)$ and $(3, 5)$, respectively, is $(x-2)^2 + \left(y - \frac{7}{2}\right)^2 = \frac{13}{4}$
- (ii) Show that the line $3x + y = 2$ does not touch nor cross the equation **3**
given in part b (i).
- (c) Find any points of inflexion on the curve $y = x^4 - x^2 + 1$ **3**
- (d) Consider the function $y = \sqrt[3]{x^4} - 1$.
- (i) Show that it has only one stationary point. **2**
- (ii) Test the nature of this stationary point. **2**
- (iii) Draw a neat sketch of this function. **1**

THE END

year 11 Ext ① 2008 T1 exam

Q1

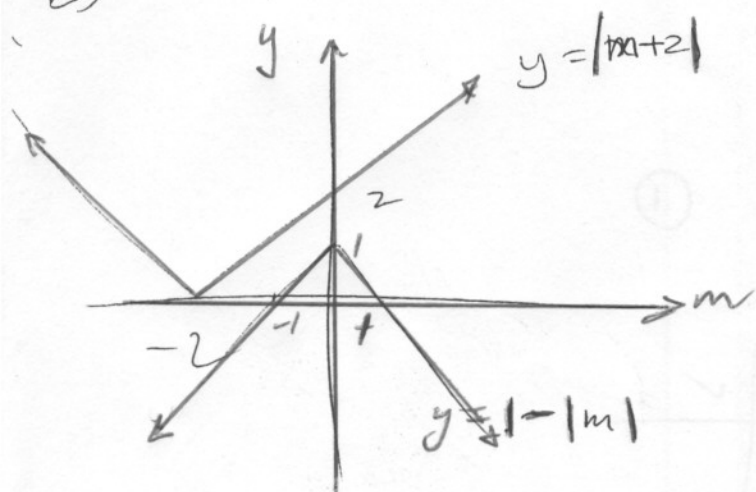
(a) $5^x \times 25^{2x+1} = 125^x$

$\therefore x + 4x + 2 = 3x$ ①

$2x = -2$

$x = -1$ ①

(b)



all real m .

$$\begin{aligned}
 (c) \quad \sqrt{2} \tan 60^\circ - \frac{\cos 30^\circ}{\sin 45^\circ} &= \sqrt{2} \times \frac{\sqrt{3}}{2} - \frac{\frac{\sqrt{3}}{2}}{\frac{1}{\sqrt{2}}} \quad (1) \\
 &= \frac{\sqrt{3}}{\sqrt{2}} - \frac{\sqrt{3}}{2\sqrt{2}} \times \sqrt{2} \quad (1) \\
 &= 0.
 \end{aligned}$$

$$(d) \quad x^2 + 3xy - 4y^2 = 0.$$

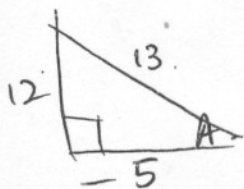
$$(x + 4y)(x - y) = 0.$$

$$x = -4y \quad \text{or} \quad x = y$$

$$\left(\frac{x}{y} = -4 \right) \quad \therefore \frac{x}{y} = 1. \quad (1)$$

$$(e) \quad \tan A = -\frac{12}{5}.$$

Second Q as $\sin A > 0$.



$$\begin{aligned}
 \sec A &= \frac{1}{\cos A} \\
 &= -\frac{13}{5}. \quad (1)
 \end{aligned}$$

$$(f) \quad (i) \quad \frac{d}{dx} (3\sqrt{\pi} - 2x^5) = -10x^4 \quad (1)$$

$$(ii) \quad \frac{d}{dx} \sin x \cos x =$$

$$= \sin x (-\sin x) + \cos x (\cos x)$$

$$= \cos^2 x - \sin^2 x \quad (1)$$

$$\text{or} \quad 1 - 2\sin^2 x$$

$$\text{or} \quad 2\cos^2 x - 1$$

$$(f)(iii) \frac{d}{dx} [\ln 2x]^{\frac{7}{2}} = \frac{7}{2} [\ln 2x]^{\frac{7}{2}-1} \times \frac{1}{x}$$

$$= \frac{7}{2x} [\ln 2x]^{\frac{5}{2}} \quad (2) \quad (3)$$

$$(iv) \frac{d}{dx} \left(\frac{2x}{4x+1} \right) = \frac{(4x+1)(2) - (2x)(4)}{(4x+1)^2}$$

$$= \frac{2}{(4x+1)^2} \quad (1) \quad (1)$$

Q2.

$$(a) \log_{10} y = \frac{2}{3} \log_{10} x - 2$$

$$= \log_{10} x^{\frac{2}{3}} - 2.$$

$$= \log_{10} x^{\frac{2}{3}} - \log_{10} 100 \quad (1)$$

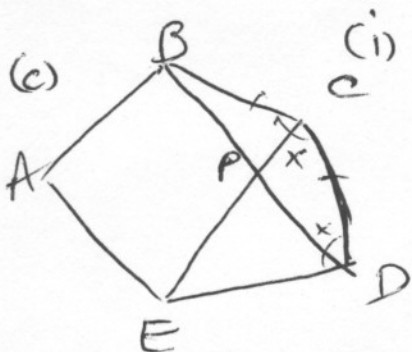
$$= \log_{10} \left(\frac{x^{\frac{2}{3}}}{100} \right) \quad (1)$$

$$\therefore y = \frac{x^{\frac{2}{3}}}{100} \quad (1)$$

$$(b) y = e^{x^2}$$

$$y' = 2xe^{x^2} \quad (1)$$

$$\therefore \text{when } x=1, y' = 2e. \quad (1)$$



(i) In $\triangle CDE$ & $\triangle BCD$.

$BC = CD$ (ABCDE is a regular pentagon)

$\angle BCD = \angle CDE$ (equal angles of a regular pentagon).

$CD = ED$ (equal sides of regular pentagon)

$\therefore \triangle CDE \cong \triangle BCD$ (SAS) 3.

(ii) $\angle ECD = \angle CDB$ (corresponding angles of congruent triangles equal).

$\therefore PC = PD$ (equal sides opposite equal angles in $\triangle CDP$)

2

c) (ii) $BD = CE$ (corresponding sides of congruent triangles equal) $\left(\frac{1}{2}\right)$ (4)

$$PD = PC \text{ (from ii)}$$

$$\therefore BD - PD = CE - CP$$

$$\therefore BP = PE \left(\frac{1}{2}\right)$$

since $AB = AE$ (sides of regular pentagon) $\left(\frac{1}{2}\right)$

$\therefore ABPE$ is a kite (two pairs of adjacent sides equal) $\left(\frac{1}{2}\right)$

d) $f(x) = \sqrt{x+2}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h+2} - \sqrt{x+2}}{h} \quad (1)$$

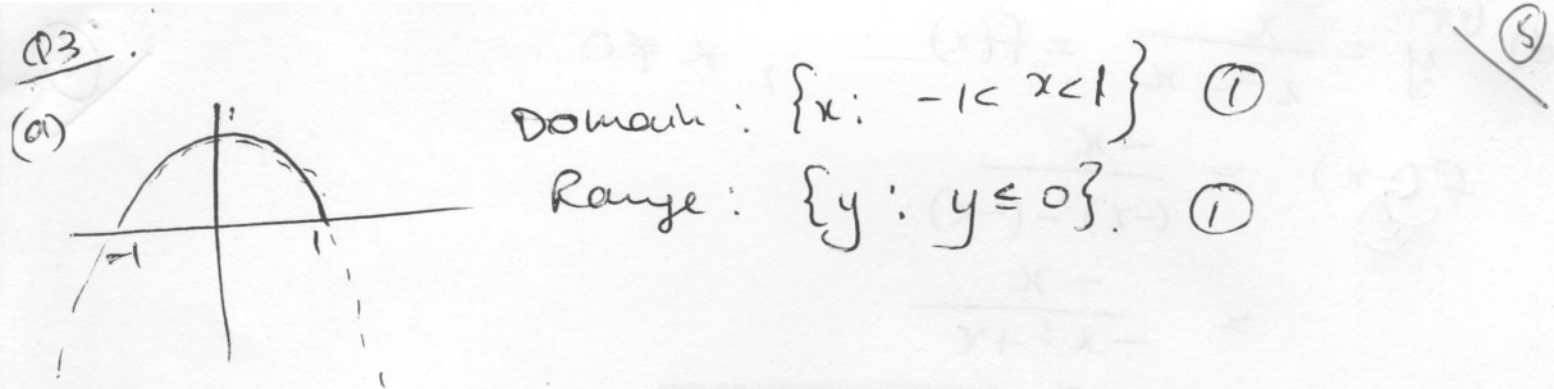
$$= \lim_{h \rightarrow 0} \frac{\sqrt{x+h+2} - \sqrt{x+2}}{h} \times \frac{\sqrt{x+h+2} + \sqrt{x+2}}{\sqrt{x+h+2} + \sqrt{x+2}}$$

$$= \lim_{h \rightarrow 0} \frac{x+h+2 - (x+2)}{h(\sqrt{x+h+2} + \sqrt{x+2})}$$

$$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h+2} + \sqrt{x+2}} \quad (1)$$

$$= \frac{1}{2\sqrt{x+2}} \quad (1)$$





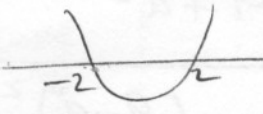
(b) $y = x^3 - 12x$

$y' = 3x^2 - 12$ ①

$y' > 0$ when $3x^2 > 12$

$x^2 > 4$

$\therefore \{x: x < -2 \cup x > 2\}$ ①



(c) $\lim_{x \rightarrow 0} \frac{2 \sin x}{\tan(\frac{x}{2})} = \lim_{x \rightarrow 0} \frac{4 \frac{\sin(\frac{x}{2}) \cos(\frac{x}{2})}{\frac{\sin(\frac{x}{2})}{\cos(\frac{x}{2})}}}{\frac{\sin(\frac{x}{2})}{\cos(\frac{x}{2})}}$ ①

$= \lim_{x \rightarrow 0} 4 \cos^2(\frac{x}{2})$

$= 4$ ①

Alternatively: $\lim_{x \rightarrow 0} 2 \sin x \times \frac{1}{\tan(\frac{x}{2})}$

$= \lim_{x \rightarrow 0} 2 \frac{\sin x}{x} \times \frac{x}{\frac{x}{2}} \times \frac{1}{\frac{x}{2}} \times \frac{\frac{x}{2}}{\tan \frac{x}{2}}$

$= 2 \times 1 \times x \times \frac{2}{x} \times 1$

$= 4$

$$c) \quad x^3 - x, \quad x \neq 0.$$

$$\begin{aligned} f(-x) &= \frac{-x}{(-x)^3 - (-x)} \\ &= \frac{-x}{-x^3 + x} \\ &= \frac{x}{x^3 - x} \\ &= f(x) \quad \therefore \text{even.} \end{aligned}$$

②

$$e) \quad \cos x = \frac{9-a^2}{9+a^2}$$

$$\sin x = \pm \sqrt{1 - \frac{(9-a^2)^2}{(9+a^2)^2}}$$

(using $\sin^2 x + \cos^2 x = 1$)

①

$$= \pm \frac{\sqrt{(9+a^2)^2 - (9-a^2)^2}}{9+a^2}$$

$$= \pm \frac{\sqrt{81 + 18a^2 + a^4 - 81 + 18a^2 - a^4}}{9+a^2}$$

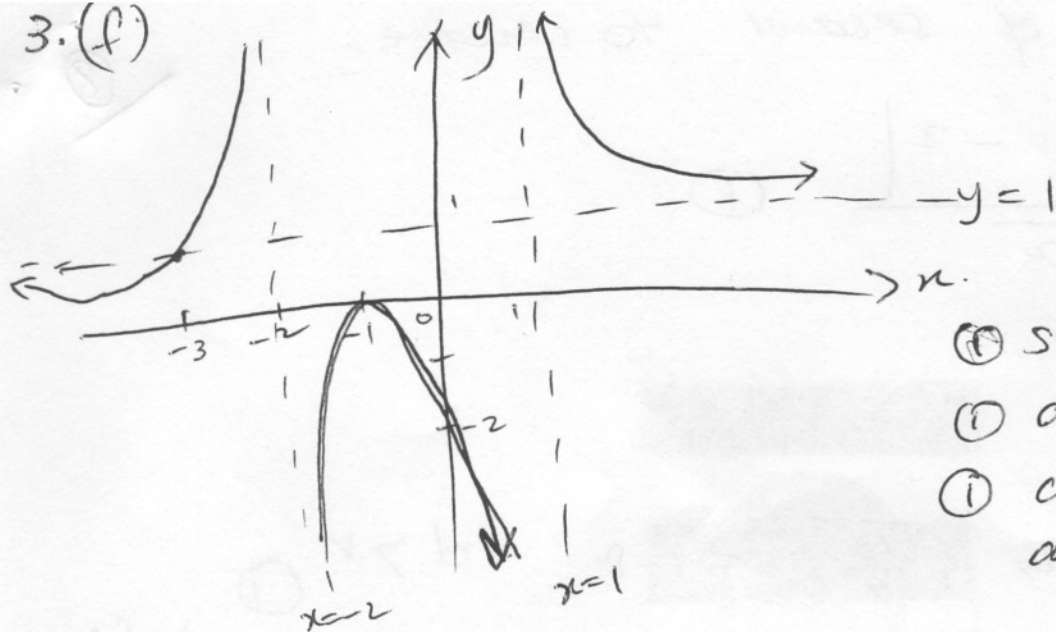
$$= \pm \frac{\sqrt{36a^2}}{9+a^2}$$

$$= \pm \frac{6a}{9+a^2}$$

①

3. (f)

⑦



① shape

① asymptotes

① critical pt $x = -3$
and intercepts.(g) $A(3, -7)$ $B(8, 2)$
5:3

$$P = \left[\frac{9+40}{8}, \frac{-21+10}{8} \right] = \left(6\frac{1}{8}, -1\frac{3}{8} \right)$$

①
correct formula.

Q4.

(a) $2x = 0.1$ or 3.04

$x = 0.05$ or 1.52

$x = 0.1$ and 1.5 (1 dp). ②

(b) (i) $\frac{y-2}{x-1} \times \frac{y-5}{x-3} = -1$ ①

$\therefore y^2 - 7y + 10 = -(x^2 - 4x + 3)$

$x^2 + 4x + 4 + y^2 - 7y + \frac{49}{4} = -13 + 4 + \frac{49}{4}$ ①

$\therefore (x+2)^2 + (y - \frac{7}{2})^2 = \frac{13}{4}$

4(b) (ii) Distance of secant to centre.

$$d = \frac{|3 \times 2 + \frac{7}{2} - 2|}{\sqrt{3^2 + 1^2}} \quad (1)$$

$$= \frac{15}{2\sqrt{10}}$$

(1)

$$\approx 2.37$$

now radius is $\sqrt{\frac{13}{4}} \approx 1.8$ $\therefore d > r$ (1)

$\therefore$ line doesn't touch nor cross the circle.

(c) $y = x^4 - x^2 + 1$

$$y' = 4x^3 - 2x \quad (1)$$

$$y'' = 12x^2 - 2$$

For possible points of inflexion $y'' = 0$

$$\therefore x^2 = \frac{1}{6} \quad \therefore x = \pm \frac{1}{\sqrt{6}}$$

test $x = \frac{1}{\sqrt{6}}$

$$x = -\frac{1}{\sqrt{6}}$$

x	0	$\frac{1}{\sqrt{6}}$	0.5
y''	-2	0	1

- 0 +

$\therefore$ change in concavity

x	$-\frac{1}{2}$	$-\frac{1}{\sqrt{6}}$	0
y''	1	0	-2

+ 0 -

$\therefore$ change in concavity.

$\therefore$ points of inflexion at

$$\left(\frac{1}{\sqrt{6}}, \frac{31}{36}\right) \text{ and } \left(-\frac{1}{\sqrt{6}}, \frac{31}{36}\right) \quad (1)$$

(d) $y = x^{4/3} - 1$

(i) $y' = \frac{4}{3} x^{1/3}$

For stat. pt $y' = 0 \therefore x = 0$ only

$\therefore$ stat pt at $(0, -1)$ ①

(ii) now $y'' = \frac{4}{9} x^{-2/3}$

at $x = 0, y'' = 0 \therefore$ use $\frac{dy}{dx}$ to test.

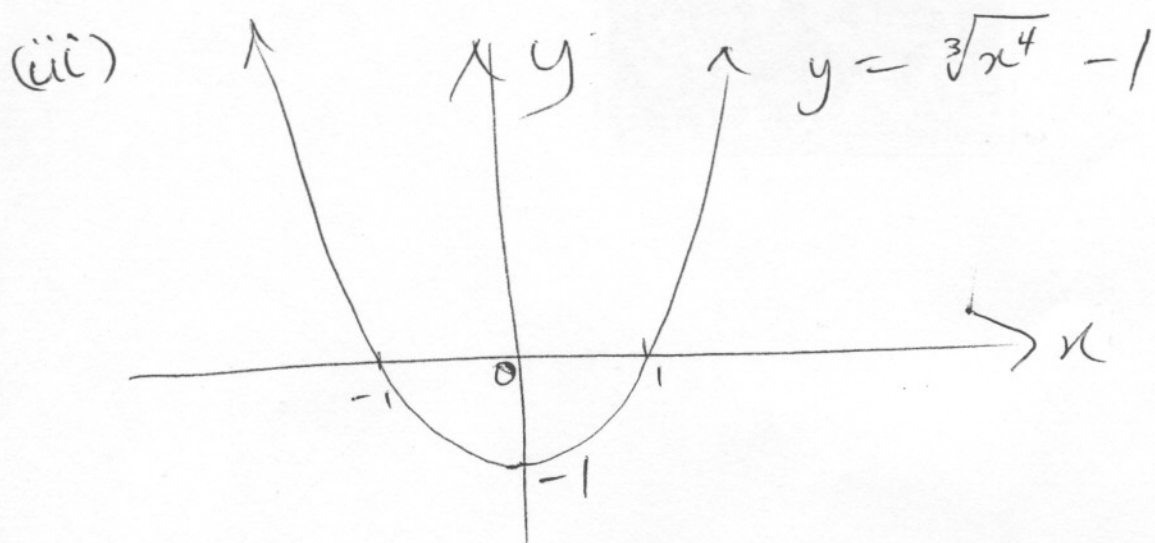
x	-0.1	0	0.1
y'	-0.62	0	0.62

①

- 0 +



$\therefore$ minimum at $(0, -1)$ ①



①